

A game for learning history on mobile devices

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Abstract This paper presents a game designed to support teaching and learning of Portuguese History to 6th grade students. Firstly, a state of the art of mobile game-based learning for History is presented. Then we describe shortly the research carried out which aimed at the analysis of the games most played by students, followed by the identification of the learning principles proposed by Paul Gee and the game mechanics to propose a game structure. Data was collected from students in the 2nd cycle of the Portuguese Educational System ($n = 508$) through a questionnaire. Finally, we describe the game, that was developed based on those findings, related to a Portuguese historical event - the Republic Implementation in 1910.

Keywords Mobile learning · 1910 mobile game · History · Game preferences survey

1 Introduction

Although it has been argued for more than a decade that ubiquitous learning will revolutionise education, it is only with the recent widespread adoption of various mobile devices in everyday life that researchers have awoken to the importance of using those devices in the educational field.

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“In recent years, mobile learning has undergone a significant transformation due to rapidly growing ownership of smartphones and tablets, accompanied by the proliferation of apps, social networks and mobile-friendly open access resources” (Kukulska-Hulme 2014, p.12).

Games, which have been applied in education for a long time, have also benefited from these advances in technology and from the portability of multiple devices (Jeong and Kim 2007). Consequently, it is very common to see today’s young people spending their free time playing on mobile devices. And students do it with an enthusiasm they do not show in other school activities. In fact, the possibility that the player may take a certain game with him/her and play it anywhere - because he/she makes the mobile phone an extension of his/her cognition - is quite attractive. Being aware that games aligned with mobile technologies can enhance motivation, we believe they can be used as a powerful mechanism for teaching formal contents, as they are fun and learning occurs in every move without the player even paying much attention (Prensky 2010).

In this paper, we will present a game for learning History, based on previous research about students’ game preferences. To inform the design, we will present the empirical study we conducted, which aimed to analyse the most commonly played games by students, drawing on the learning principles proposed by Gee (2003).

2 Using games for learning history

A game is a system that has a set of rules, obstacles and choices and the player needs to learn through his failures to be successful (Gee 2007). For Klopfer (2008), through the exploration of the game environment, players learn from their successes and failures, and receive constant feedback that helps to execute the task. They have decisions to make, they have a problem to solve, risks to take and sometimes an experience of failure, they learn how to manipulate and control complex environments and also collaborate with other players (Douch et al. 2010; Prensky 2006; Williamson 2009). Games are, for all of these reasons, systems that can provide the best environment for learning. After playing numerous games, Gee (2003) stated that “good video games incorporate quite good perspectives on how learning, thinking, and problem solving work in the world and should work in schools” (p.76). He concluded that education has much to learn with video games.

Orr and McGuinness (2014) recognized that “digital technology has facilitated this more active and social view of learning, by transforming the role of learners from passive receivers to active constructors of knowledge and by affording more participatory experiences of learning and the co-construction of knowledge” (p. 222). Students have an active role in their learning by using games. They can repeat activities several times, they can progress at their own pace, and failure is an opportunity for learning and for trying again, not something to be concerned about.

Through an analysis of research about the impact of games on learning, Orr and McGuinness (2014, p. 235) concluded that “the research reviewed confirms the potential for GBL [Game-Based Learning], to improve cognitive functioning [e. g. spatial skills, visual attention skills, problem solving] (...). The potential to improve school attainment is also confirmed, however, more so in science-based domains” (p.

235). Similarly, Hamari et al. (2016) suggest that “educational video games can indeed be an effective means of creating conditions for flow, heightened engagement (including interest, concentration, as well as enjoyment), and immersion, which can further facilitate the learning of complex strategies through game-based learning” (p. 177).

Commercial video games have been used in the discipline of History as reported by Şar (2012). Drawing on questionnaire responses from 219 students who attended 6th to 8th grades, the author said that “they mostly play games such as Call of Duty, Age of Empires, Age of Mythology, Assassin’s Creed. (...) By playing these games the students stated that they gain knowledge on Mesopotamia, Rome, Eastern Culture, Byzantium, Ottoman and Silk Road” (Şar 2012, p. 779). Lucas and Sherry (2004) in their study found that the male students play these type of games, that focus on the theme of war, more often, which also attract more male players than female ones. Importantly Şar (2012) noted that “the students who reported to play history themed non-educational computer games [were] also found to be more enthusiastic to visit museums or other historical places” (p. 779). These games create curiosity in the students about the history content.

Squire (2011) used a game named Civilization (Civ) in his history lessons and noted “that Civ (1) provides a framework tying together world geopolitical history; (2) gives fluency with a model that is useful for analyzing world history on broad time scales; and (3) inspires revisiting, tinkering, and socializing. These same factors make good success criteria for games more generally” (p. 28). Civ was a video game that gave students the possibility to experience the evolution of the human history through the wars between the different civilizations. Squire argued that the students frequently searched for information on historical facts that helped them to achieve a better performance in the game. They were engaged and they understood the facts behind some situations that occurred in human history, because the game responded to the player’s decisions about politics, science progress, and materials available in the region.

Several research projects have been conducted to develop games that engage students to learn some facts about history content. In Italy, an excursion-game for cellular phones was developed for an archaeological park. In the game, students in groups explore the park identifying places from clues provided. The evaluation of the game confirmed that the “game is indeed capable to engage students and to permit an effective and exciting visit of archaeological sites, thus enhancing the overall students’ experience” (Ardito et al. 2007, p. 365).

Two universities in the Czech Republic developed a “project [named] *Stories from the History of Czechoslovakia*, a serious game for teaching history of the Czech lands in the 20th century” (Šisler et al. 2012, p. 67). The game was divided into five periods of the history of Czechoslovakia. The content is based on testimonies of eyewitnesses of each of these periods for constructing realistic and appealing narratives. It is also possible to access a multimedia encyclopaedia to access additional information. Šisler and colleagues conducted a study and found that the majority of students were mildly sceptical about the use of digital games for learning. They state that games should be used as a supplement and the teacher has to facilitate student group discussions about their experiences. The authors argue that “the game should be designed from the beginning with the idea that it will be later integrated within the formal schooling system. Designing supplementary activities and supporting teachers is vital” (idem, p.74).

In the Universiti Sains Malaysia a game was developed for learning historical content on mobile devices. The name of the project was M-History. It was about a legendary warrior, Merong Mahawangsa. The game was divided in five moments where each player had a mini game to play and then had to answer questions relating to the school curriculum. They chose to implement a RPG (Role Playing Game) based on previous research that suggests that it is the best environment to learn history (Lee et al. 2014).

It is important to notice that the results achieved in different projects are similar. Students effectively learn history from the experience of the game and their curiosity increases leading them to search for further information. It is also evident that, for a good integration of the game in the discipline content, it is important to prepare activities and materials to support the teacher's use of the game in the classroom.

Museums are also developing serious games to engage students in history content with successful results. The project Playhist in Carnegie Mellon University (USA) uses 3D technology to immerse the student in the environment. The player has to play some mini games to progress in the game (Perez-Valle et al. 2014).

Looking at these projects makes us think about all that is needed to develop a game. Many teachers experience several types of barriers when they use games to teach, including: expertise (they need training), systemic (it is difficult to integrate game playing in the rigid curriculum), financial (the cost of games) and technical (lack of dedicated equipment in school) (van Rosmalen et al. 2014). The use of mobile devices by students will address some of these barriers but they can use them anytime and anywhere, playing the game inside and outside the classroom. It is also important for teachers to follow the progress of the game of every student.

For Orr and McGuinness (2014), "it is vital that the learning promoted through a game is appropriately transferable to both the curriculum goals of the classroom and other real world contexts" (p. 235). Only when this is possible will the game be adopted by the teachers and students because then it will be purposeful and warrant the effort required by both parties. When developing a game, it is therefore important to think about not only the student's engagement, but also to create the tools and resources that will help teachers maximize the use of the game for learning purposes.

However, Beland and Murphy (2015) refreshed the debate about the potential of mobile devices for personal use to cause distraction when they found that it can have negative effects on low-achieving students. The authors concluded that "banning mobile phones improves outcomes for the low-achieving students (...) the most and has no significant impact on high achievers" (p. 17). It is important to notice that they do not question the possible benefits of the mobile devices for a structured learning process, only the personal use by students. Indeed, when using them in the class "[t]he most important (...) are the teaching strategies and methods created with and for the technology" (Moura and Carvalho 2013, p. 68).

This suggests that being aware of the games that students already play provides us with some clues about what they like and what will be more engaging for them. When developing a history game, it is important to interact in a team of "game designers, programmers, and artists, but also historians and educationalists" (Šisler et al. 2012, p. 73), which make it possible to achieve educational goals with the expertise of game developers and to overcome the identified barriers (van Rosmalen et al. 2014).

3 Research

An ongoing project conducted in Portugal, named “From Games to Interactive Activities for Mobile Learning”, aims to identify the mobile games that Portuguese students play most, especially on mobile devices. Students from the 5th grade of Basic Education up to Master’s degree level completed a survey, with a sample size of 2303 players of mobile games. Overall, results from the study indicated that i) students’ preferences change according to their educational level and gender, ii) female students show more interest in casual and simple games, while male students prefer games that involve planning, strategy and cooperative work, iii) female students prefer to play alone, while male students prefer to play in online communities and iv) older players value the games with impressive graphic effects that engage them in virtual environments (Carvalho and Araújo 2014).

In this paper, data obtained from 5th and 6th grade students ($N = 508$), aged 10 to 12, will be presented and their game preferences will be analysed to inform the design of the “1910” mobile game.

3.1 The games most played by students aged 10–12

The first phase of the project aimed to identify the games most played, as well as students’ game habits on mobile devices. To achieve these purposes, an online survey was carried out (Babbie 1997) in Portugal between May and November 2013. The survey explored four dimensions, namely i) Characterisation of the students; ii) Characterisation of their game habits; iii) Preferences about games and iv) Preferences for learning school contents through games. The survey was validated by experts in the field and approved by the National Board of Education (Direção Geral de Educação) of Portugal. All schools were invited to participate in the survey online through the website of the agency belonging to the Ministry of Education and by email. The survey was anonymous and completed by students online, most of the respondents were from the north of Portugal, and the sample is representative in relation to the target population (see Table 1). It is likely that the sample includes non-response bias due to school and student self-selection to participate.

3.1.1 Sample characterisation

Out of the 649 respondents, 508 played games on mobile devices, of whom 58.7 % were male and 41.3 % were female (see Table 1), with an average age of 11.2 years. In this study, we corroborate findings related to players’ gender already presented in other

Table 1 Number of answers obtained from students aged 10–12

Educational Level	Respondents				Respondents playing games			
	Total	Gender	f	%	Total	Gender	f	%
5th and 6th grade	649	Male	367	56.5	508	Male	298	58.7
		Female	282	43.5		Female	210	41.3

studies (Lenhart et al. 2008, Lucas and Sherry 2004; Simons et al. 2012) indicating that male students are more likely to play than female students (see Table 1).

When asked about the mobile devices on which they play, the responses in order of popularity were: a laptop (78.7 %), a mobile phone (63.8 %), a PSP (handheld game console) (46.7 %), a tablet (44.7 %), a Nintendo 3DS (handheld game console) (24.2 %) and a smartphone (21.3 %). Regarding the mobile device they used most to play their favourite game, the most popular was a laptop (45.3 %), followed by a mobile phone (14.4 %), a tablet (13.4 %), a PSP (13.4 %), a smartphone (5.1 %) and a Nintendo 3DS (2.4 %). 6.0 % did not state which device they used to play their favourite game.

3.1.2 Game habits

When asked about the time they devoted to games, 41.1 % of the respondents indicated a range of 1 to 5 h a week, 34.3 % less than an hour and 13.4 % between 6 to 10 h. 4.7 % of the respondents indicated 11 to 20 h and also 4.7 % indicated more than 20 h per week (1.8 % of the respondents did not answer this question). On average, respondents played 4.4 h per week; male respondents played an average of 5.5 h and their female peers 2.8 h.

With regards to the respondents' preference for game partners, the majority of the respondents indicated that they preferred to play alone (57.1 %). Those who preferred to play with others online (42.9 %) were distributed as follows: 27.4 % with friends and/or acquaintances, 23.8 % with colleagues, 13.4 % with siblings, 12.4 % with other family relatives, 10.0 % with strangers and 2.6 % with parents.

3.1.3 Preferences about games

Taking into account the great diversity of the games identified (141 games) by the respondents (Carvalho et al. 2014), only the five most common games indicated were analysed.

The five most played games were identified by 31.7 % of the survey sample reported here. Data highlight that the Pro Evolution Soccer option was only identified by the male students and that Pou and Subway Surfers were predominantly female students' options (see Table 2).

Table 2 Top 5 favourite games, distribution by gender ($n = 161$)

Game	Male ($n = 119$)		Female ($n = 42$)	
	f	%	f	%
Grand theft auto (GTA)	39	32.8	1	2.4
Pro evolution soccer (PES)	26	21.8	0	0.0
Subway surfers	16	13.4	15	35.7
Minecraft	17	14.3	5	2.4
Counter strike (CS)	15	12.6	3	7.1
Pou	6	5.1	22	52.4

As our sample has more responses from males (58.7 %) than from females (41.3 %), typically male games stand out, characterised as violent (GTA, CS), sport (PES) and sandbox (Minecraft) (Terlecki et al. 2010) (see Table 2). These highly competitive games require effort and take several hours to complete. It is important to note that some of the games mentioned are not suitably age rated for the students that are part of our sample, whose preferences are therefore, at variance with the European standards (PEGI 16 and 18). In addition, 2.3 % of the boys aged 10–12 regularly play games rated 16, and 23.2 % of male and 4.3 % of female regularly play games rated 18+ (PEGI18) (Carvalho et al. 2014).

This data indicates that the major differences can be found in the gender-related preferences. While male students prefer longer games, which sometimes involve group cooperation, female students prefer faster games, which they can play alone without interacting with other players (Carvalho et al. 2014, p.30). We found that the students also have different preferences in relation to the themes of the games (see Table 3). The boys prefer action and adventure (GTA), war (CS) or football (PES, FIFA) games, while girls prefer games about everyday life, where they can take care of an “animal” (Pou), people (Sims) or experience the fashion world, creating and selling clothes (Stardoll). They can also test some skills such as speed of reaction (Super Mario and Subway Surfers).

Sport games, mainly football, are a typical boys’ activity (Blakemore et al. 2008; Cherney and London 2006; Sin et al. 2014; Terlecki et al. 2010; Williams et al. 2009). For this reason it is understandable that in the male list of the most played games there are football games. On the other hand, adventure and the ability to build something are very appealing ingredients for male players (Blakemore et al. 2008), like the game Minecraft, which appears in 3rd place in Table 3.

Casual games like Pou and Subway Surfers are more appealing for female players. These are games which are quick to play, and highly rewarding without major punishments; their aim is to collect as many coins as possible to purchase items for avatars (e.g., Pou and Subway Surfers). Some players like to share their purchases and their achievements in the game through Facebook, SMS or email. This social interaction is mentioned by Terlecki et al. (2010) as important for female players.

In the questionnaire the students ($n = 508$) were asked to express their opinion about the difficulty level of the games they played most. The most popular response was ‘Moderate’ (40.8 %), followed by ‘Easy’ (22.4 %) and ‘Very Easy’ (16.9 %). Fewer

Table 3 Top 5 favourite games according to gender ($n = 208$)

Ranking	Male ($n = 128$)	Female ($n = 80$)
1st	Grand theft auto (<i>Rockstar Games</i>) (30.5 %)	Pou (<i>Zakeh</i>) (27.5 %)
2nd	Pro evolution soccer (<i>Konami</i>) (20.3 %)	Super mario (<i>Nintendo</i>) (20.0 %)
3rd	Minecraft (<i>Mojang</i>) (13.3 %)	Subway surfers (<i>Kiloo Games & Sybo Games</i>) (18.8 %)
4th	Subway surfers (<i>Kiloo Games & Sybo Games</i>) (12.5 %)	The sims (<i>Electronic Arts</i>) (17.5 %)
5th	Counter strike (<i>Valve Software</i>) (11.7 %)	Stardoll (<i>Stardoll</i>) (16.3 %)
	FIFA (<i>EA Sports</i>) (11.7 %)	

students indicated that the games they played the most were either ‘Hard’ (12.2 %) or ‘Very Hard’ (7.7 %).

Students were also asked to indicate the importance of some characteristics of the game for keeping them playing. The differences between male and female students are noteworthy, as we can see in Table 4.

We, therefore, conclude that male students particularly value the aesthetic of the game, the progression and the competition while female students most appreciated the aesthetics of the game (characters, graphic effects and animations) and opportunities for progression in the game as the characteristics that make students keep playing.

3.1.4 Learning course content through games: students’ preferences

Students were asked if they would like to use games to learn course content and most of them answered positively (84.6 %). These students ($n = 430$) were asked to indicate the type of game they would like to use in class. Adventure (66.0 %), Action (59.1 %) and Sport (50.5 %) were the types most commonly chosen, notably similar to games most played outside school.

4 Learning principles identified in the games most played

Based on the 36 learning principles identified by Gee (2003), the five games most played by all students (irrespective of gender) were analysed. Eight of these principles were identified: i) *Psychosocial Moratorium* (“Learners can take risks in a space where real-world consequences are lowered” (p.67); ii) *Committed Learning Principle* (“For a little input, learners get a lot of output” p.67); iii) *Amplification of Input Principle* (“Learners participate in an extended engagement (lots of effort and practice) as an extension of their real-world identities in relation to a virtual identity to which they feel some commitment and a virtual world that they find compelling” p.67); iv) *Achievement Principle* (“For learners of all levels of skill there are intrinsic rewards

Table 4 Characteristics of the game for keeping the player playing

Characteristics of the game for keeping playing	Male ($n = 298$)	Female ($n = 210$)
Graphic effects and animations	79.9 %	70.0 %
Gameplay	78.2 %	59.9 %
Characters	74.2 %	79.5 %
Being able to play with others	72.8 %	52.9 %
Being able to improve one’s score	71.1 %	70.0 %
The game having many levels	70.8 %	72.4 %
Scenarios	69.5 %	49.5 %
Being a long game	67.8 %	51.9 %
Friends playing the same game	64.4 %	55.2 %
Being able to play online with others	63.4 %	33.3 %
Storytelling	57.0 %	58.6 %
Sounds	55.7 %	42.9 %

from the beginning, customized to each learner's level, effort, and growing mastery and signaling the learner's ongoing achievements" p.67); v) *Practice Principle* ("Learners get lots and lots of practice in a context where the practice is not boring (i.e. in a virtual world that is compelling to learners on their own terms and where the learners experience ongoing success). They spend lots of time on task" p. 71); vi) *Regime of Competence* ("The learner gets ample opportunity to operate within, but at the outer edge of, his or her resources, so that at those points things are felt as challenging but not "undoable" p. 71); vii) *Intuitive Knowledge* ("Intuitive or tacit knowledge built up in repeated practice and experience, often in association with an affinity group, counts a great deal and is honored. Not just verbal and conscious knowledge is rewarded" p.111) and viii) *Affinity Group Principle* ("Learners constitute an "affinity group", that is, a group that is bonded primarily through shared endeavors, goals, and practices and not shared race, gender, nation, ethnicity, or culture" p.197).

The reasons mentioned by the students for liking the games most played are the opportunities to do violent actions, to drive a car or to fight criminals. These dangerous and morally reprehensible activities can be carried out in a safe virtual environment where it is possible to take risks without real consequences (*Psychosocial Moratorium* - according to Gee 2003). Due to this, it is possible to test hypotheses and learn through the results achieved. In all the games, we also identified the *Amplification of Input Principle* (Gee 2003), because the tasks are set in a simple way that produces feedback without a need to actually do them as in the real world. For instance, a football match can be completed between 4 and 12 min and not in 90 min as in the real world. These two principles are present in the world of the game creating an environment of more engagement.

When analysing common characteristics in these games, we can identify that in all of them it is necessary to fulfil tasks: these can be repetitive (Pou, PES, Subway Surfers) or concern the completion of missions (GTA, CS, Minecraft). Most of these tasks are feasible in short time-periods and players are rewarded on successful completion. Carrying out these tasks in a regular manner makes it possible to progress in the game and to develop the avatar (all games listed except sport) by customising it or acquiring items. It is the *Committed Learning Principle* (Gee 2003) when students apply their time on a regular basis to doing the tasks that make the avatar development possible, thus feeling a commitment towards this virtual identity.

Some of the tasks are repetitive and this makes it possible to improve the player's capabilities (*Practice Principle*; *Regime of Competence*; *Intuitive Knowledge* - Gee 2003). The players can access tutorials to explain the game and new types of activity that come to light as they progress. Through repetition players can practice similar activities in the game thus improving his/her capabilities, for example by driving vehicles in GTA, by playing soccer matches in PES or by running on a railway line in Subway Surfers (*Practice Principle*). Even if the player cannot complete a task at the first attempt, he/she feels that he/she is capable of accomplishing it by trying again and again, improving his/her performance (*Regime of Competence*). For instance in Subway Surfers, when a player helps Jake, the avatar, to run on a railway line jumping and moving between the tracks in every attempt he/she can go further and feel that he/she is capable of doing even better the next time around. Also every action made in the

game counts and the player is rewarded for that even when he/she loses the match (*Achievement Principle*). For example, in Subway Surfers the player accumulates all coins caught and in each attempt the difficulty increases. Practice and experience allow the player an intuitive and tacit knowledge that is highly valued between players who share an interest in the same game (*Intuitive Knowledge*), for instance peers talk about the achievements in the game and recognize the effort need.

Another common characteristic in the most popular games is social interaction with other players - the *Affinity Group Principle* (Gee 2003) Through sharing items (Pou, Subway Surfers) or by using a multiplayer mode (GTA, PES, CS and Minecraft) the players share the same interest in the same game, enjoy sharing their achievements because the other players understand and value them. By talking about their achievements and discoveries it is possible to learn from the experience of other players or to teach others.

5 Game: 1910

The findings from the survey of students concerning their game preferences, the game analysis and the identification of learning principles guided us during the design of the mobile game “1910”. Knowing the students preferences and what they play, helped us to understand what can be more engaging for them from all the possibilities available, as Salen and Zimmerman (2004) reported, to design a game. All learning principles identified were applied to the game except Practice principle, due to the characteristics of the game content that will be presented.

5.1 Planning

The game has been designed according to two prerequisites: i) to allow the formal learning of contents, and ii) to be a creative game that will attract students’ attention in order to help them learn.

We decided to make a game where the player can live through history. This required a big effort on the part of the design team in order to develop all elements of the storytelling. We decided to use photographs from historic events to create a world. In this world, it is possible to experience historical events in a safe way (Psychosocial Moratorium Principle), also it is possible to go through several years of history experiencing only the important moments (Amplification of Input Principle). The avatar needed to be a fictionalized character that could be involved in the historical events and with whom the player could identify with (Committed Learning Principle). The reward system has historical significance as the coins belong to the 19th century. Within the events in the game, some quests have to be completed. These quests demand students’ understanding about the historical events. Students may improve their results through repeating tasks (Achievement Principle and Regime of Competence). A leaderboard, where only the top five scores of the classmates appear, will motivate students to focus their efforts, score highly and be recognized by their peers (Intuitive Knowledge). Talking about the game with their classmates will help in discovering ways to improve their achievements in the game (Affinity Group Principle).

5.2 Gameplay

This game was developed for Android system, as this is the mobile operating system most commonly used by the Portuguese population.

To interact with the game, the player needs to:

- a) touch the screen to select the correct answer or
- b) use the drag and drop function (e.g., choosing the right words or the most impressive pictures).

When the player starts the game, he/she sees how a journalist lived from 1890 to 1911. The player must help him to write the most informative news to be published in the newspaper, by selecting the correct words or choosing the most impressive photos. The game can be regarded as a mission, i.e., helping a journalist to write about the historical events of the republican revolution in Portugal.

5.3 Subject theme and course content

The History discipline was chosen for the setting of the game because it is an important discipline and also reflects the interest and knowledge in the project team. The course content selected was the Republic Implantation in Portugal and the game was designed for the 6th grade (students aged 11–12).

The designed game can be used in two different ways. The students can play the game in the classroom or outside school (e.g., for homework) and then discuss the outcomes in class. The teacher can access the students' results by checking the report in an Excel page.

5.3.1 Theme

In the late 19th century, the Portuguese monarchy faced great difficulties. The Republican Party took advantage of the discredit of the monarchy to attempt to overthrow it. To achieve this purpose, the leaders of that party organized several riots. In 1908, King Carlos I and his heir-apparent, Prince Royal Luís Filipe, were assassinated by revolutionaries. D. Manuel became king of Portugal, struggling with several difficulties. However, he failed to solve the serious economic, social and political crisis that was raging in the kingdom. On 5th October 1910, a revolution led to the establishment of a republic in Portugal. After the revolution, a provisional government was created, new national symbols were adopted, elections were held and a constitution was drawn up. The governments of the 1st Portuguese Republic also took several actions to develop education and tackle the difficulties faced by workers. In spite of having failed to solve the kingdom's problems, this historical event is remarkable and is regarded as a crucial aspect to make children and young people in Portugal aware of the need to value the History of their country, as a way of preserving national memory and identity.

5.3.2 Game description

The students log into the game. The first time they log in, they watch a video that will present the role they are to play in the game - a journalist in the year of 1890. This

journalist's job is to investigate facts/events in order to write news for the newspaper he works for in Lisbon. The game starts when the journalist is in the newspaper office and he is asked to write news for the next day's edition. The director tells him that the more the newspaper sells, the higher his salary will be (see Fig. 1).

The tasks required are distributed by episodes within specific dates of a historical period (see Fig. 2).

These episodes will require greater involvement of the student to deal with the increased complexity of the tasks (see Fig. 3). As soon as the tasks are accomplished, it allows the player to unlock the next historical event in the timeline menu (see Fig. 2).

For each correct answer, the player is awarded a specific number of points. Each task has a different number of points that can be awarded based on the effort required to solve the game. So more points are awarded for completing the most difficult tasks. If the player responds incorrectly he loses half of the points of that task and has to do it again until the task is completed successfully. It is also possible to repeat each episode to achieve a better score.

In order to learn more about the Republic Implantation in Portugal, several activities were developed. Below we describe the interactive activities designed in each episode as well as the challenges that students face (see Table 5).



Fig. 1 Introduction to the game



Fig. 2 Timeline menu of historical events

It should be noted that the tasks required from the players are distributed by episodes within a historical period with notable dates (1890, 1908, 1910, 1911). These episodes require a greater involvement of the student according to the increased complexity of the tasks they have to solve. The tasks performed allow the player to unlock the time frames.

According to the reward system used, the player receives a specific number of coins for each correct answer. To pass to the next level, the student needs to have gained a fixed amount of coins. If the answer is wrong, the player does not receive the respective coin but can progress forward until the end of that episode. At the end, the player is given the chance to recover the coins he did not win, by performing the task once again.

5.3.3 Students' and teachers' roles

During the game, students and teachers perform a different role.

Student's role The game can be considered an adventure and RPG (Role Playing Game) type. After a student registers for the game and is accepted by the teacher, the



Fig. 3 Game Activities (example)

students are presented with the character that they will ‘be’, a journalist, and they are directed to a specific year where they have to complete tasks and find out what occurred in that period of time. The aim is that the students will learn through engaging with these challenges.

In that sense, in a playful way, students engage with historical events of the republican revolution in Portugal through a game designed to help them to develop several skills, including awareness of national identity.

These facts are recorded in the game notebook, giving students a summary of the major historical events studied in the subject concerned.

Teacher’s role The teacher creates a class group on the game website and accepts students’ registrations, gaining access to students results when they are playing.

Furthermore, it is the teacher who approves the access to the class. This allows the teacher to access to student progress.

6 Final remarks

Research on games about History content demonstrates that it is possible to achieve successful results and motivate students to search for more information. Playing a game gives students a purpose to understand what happened in the past and simultaneously to

Table 5 Game activities

Year	Episodes	Challenges
1890	1	The player has to listen to a set of dialogues that represent the needs and complaints of people and register the most important words in a notebook. The player (journalist) sees a news article with some blank spaces that need to be filled. To complete the article, he/she must use the notes he/she has taken and fill in the gaps with historical information.
1908	1	The player (journalist) is supposed to cover photographically the arrival of the royal family when the regicide happens. The player must choose one of the photographs to illustrate the news article for the next day's newspaper. He/she has to put together a jigsaw puzzle to get the whole picture, because the glass of the photo was broken with during the regicide.
	2	The player is challenged to find some members of a secret society. He must go through an avenue with a code name and look for a person who is able to give him/her a password that unlocks the identity of that person. When the player identifies that person, he/she will be invited to be part of a secret organization. However, to be accepted, he/she must find other members of that organization.
	3	In a café, the journalist listens to a member of the Republican Party, who is stating his political convictions. The player will be faced with several items related to the republican ideals; he must select the correct ones.
1910	1	The journalist starts a private conversation with a republican and convinces him that only a coup could implant the republic. Then, he shares the plans of Carbonaria (the secret society) with him. The player's task is to drag the names of the places to the correct places in the map. When he/she does it correctly, he/she will be provided with some historical information about each place.
	2	The player is faced with a map displaying the most important places during the revolution. The player must click on the images of those places and read the information provided. At the end, the players complete a quiz about these key points.
	3	The journalist must write a new article to inform the country about the republican victory and the new national symbols (anthem, currency, flag). The player will see the national flag uncoloured. Then, he/he must use a brush to tick the respective colours in the flag (some historical explanations will be provided). Then, the player will see the musical score of the Portuguese anthem and he/she is required to drag the title and the verses of the anthem to the correct spaces. At the end, the player listens to the chorus of the anthem. As soon as the player completes the task, he/she will be able to see the coin counter and to notice that the "real" was replaced by the "escudo".
1911	1	The journalist (years later) organizes a news article to present the measures taken by the republicans. Several sentences are presented to students in categories, but in a random way. The player must match the different sentences to the categories they apply to.

achieve better scores on the game (Ardito et al. 2007; Lee et al. 2014; Šisler et al. 2012; Squire 2011).

Our project findings suggest that students are receptive to using games in course activities, to learn school subjects. They like the repetition of tasks, which allow the development of cognitive skills or motor dexterity in the student, as well as to be identified with a character in the game.

The game "1910", created for mobile devices, was developed to motivate and engage students in learning History through dialogues with characters, adventure and challenging tasks. The notebook is an important pedagogical feature of this game. It keeps all information collected through the activities, allowing the student to review the historical information in an easy way.

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